

DNS Enumeration using nslookup

Objective:

The objective of this task is to perform DNS enumeration using the nslookup tool in order to gather information about the target domain such as:

- IP Address
- Mail Servers (MX Records)
- Name Servers (NS Records)
- DNS Configuration

This information helps in understanding the target network infrastructure during the reconnaissance phase.

Command 1: Find IP Address of the Domain

Command: nslookup example.com

Explanation:

This command queries the DNS server to resolve the domain name into its corresponding IP address.

It helps identify:

1. Web server IP address
2. Hosting infrastructure
3. Target server location

Sample Output:

Server: 8.8.8.8

Address: 8.8.8.8#53

Non-authoritative answer:

Name: example.com

Address: 93.184.216.3

Analysis:

From the output:

- The DNS server used is 8.8.8.8.
- The domain example.com resolves to IP address 93.184.216.34.
- This IP address can be used for further scanning and reconnaissance activities.

Screenshot:

```
(sanika@kali)-[~]
$ nslookup facebook.com
Server:      192.168.61.2
Address:     192.168.61.2#53

Non-authoritative answer:
Name:   facebook.com
Address: 57.144.42.1
Name:   facebook.com
Address: 2a03:2880:f312:1:face:b00c:0:25de
```

Command 2: Retrieve MX (Mail Exchange) Records

Command: nslookup -type=mx example.com

Explanation:

This command retrieves the Mail Exchange (MX) records of the target domain.

MX records identify:

- Mail servers handling emails
- Email infrastructure of the organization

Sample Output:

example.com mail exchanger = 10 mail.example.com

Analysis:

From the output:

- mail.example.com is the mail server for the domain.
- Priority value 10 indicates the preference level of the mail server.
- Mail server information may help identify third-party email services and potential attack surfaces.

Screenshot:

```
(sanika@kali)-[~]
$ nslookup -type=mx facebook.com
Server:      192.168.61.2
Address:     192.168.61.2#53

Non-authoritative answer:
facebook.com mail exchanger = 10 smtpin.vvv.facebook.com.

Authoritative answers can be found from:
smtpin.vvv.facebook.com has AAAA address 2a03:2880:11ff:fffd:face:b00c:0:686e
smtpin.vvv.facebook.com has AAAA address 64:ff9b::42dc:95fb
```

Command 3: Retrieve NS (Name Server) Records

Command: nslookup -type=ns example.com

Explanation:

This command retrieves the authoritative Name Server (NS) records of the target domain.

NS records identify:

- DNS servers responsible for the domain
- DNS hosting infrastructure

Sample Output

example.com nameserver = ns1.example.com

example.com nameserver = ns2.example.com

Analysis

From the output:

- The domain uses:
 - ns1.example.com
 - ns2.example.com
- These servers manage DNS resolution for the domain.
- Name server information helps map the DNS architecture of the organization.

Screenshot:

```
(sanika@kali)-[~]
└─$ nslookup -type=ns facebook.com
Server:      192.168.61.2
Address:     192.168.61.2#53

Non-authoritative answer:
facebook.com nameserver = a.ns.facebook.com.
facebook.com nameserver = c.ns.facebook.com.
facebook.com nameserver = b.ns.facebook.com.
facebook.com nameserver = d.ns.facebook.com.

Authoritative answers can be found from:
c.ns.facebook.com has AAAA address 2a03:2880:f1fc:c:face:b00c:0:35
c.ns.facebook.com has AAAA address 64:ff9b:b959:da0c
b.ns.facebook.com has AAAA address 2a03:2880:f0fd:c:face:b00c:0:35
b.ns.facebook.com has AAAA address 64:ff9b:8186:1f0c
d.ns.facebook.com has AAAA address 2a03:2880:f1fd:c:face:b00c:0:35
d.ns.facebook.com has AAAA address 64:ff9b:b959:db0c
a.ns.facebook.com has AAAA address 2a03:2880:f0fc:c:face:b00c:0:35
a.ns.facebook.com has AAAA address 64:ff9b:8186:1e0c
```

Command 4: Retrieve SOA (Start of Authority) Record

Command: nslookup -type=soa example.com

Explanation:

This command retrieves the Start of Authority (SOA) record of the domain.

SOA records contain:

- Primary DNS server
- Administrator contact
- DNS zone serial number
- Refresh and retry intervals

Sample Output:

origin = ns1.example.com

mail addr = admin.example.com

serial = 2024050101

refresh = 7200

retry = 3600

expire = 1209600

Analysis:

From the output:

- ns1.example.com is the primary DNS server.
- admin.example.com represents the administrator contact.
- The serial number tracks DNS zone updates.
- Refresh and retry values define synchronization intervals between DNS servers.

```
(sanika@kali)-[~]
└─$ nslookup -type=soa facebook.com
Server:      192.168.61.2
Address:     192.168.61.2#53

Non-authoritative answer:
facebook.com
  origin = a.ns.facebook.com
  mail addr = dns.facebook.com
  serial = 4207849484
  refresh = 14400
  retry = 1800
  expire = 604800
  minimum = 300

Authoritative answers can be found from:
```